# Lab6 Requirements

1. Mount Image File，使用 df -h 確認有無 mount 成功。

```
root@ryan-HPC:/home/ryan/Downloads/images# mount -o loop,offset=$((1*512)),sizel
imit=$((65536*512)) sdcard.img /home/ryan/tmp/P1_BOOT
root@ryan-HPC:/home/ryan/Downloads/images# mount -o loop,offset=$((65537*512)),s
izelimit=$((311296*512)) sdcard.img /home/ryan/tmp/P2_ROOTFS
root@ryan-HPC:/home/ryan/Downloads/images# df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           2.0G  2.3M  2.0G   1% /run
/dev/sda6       915G  319G  550G  37% /
tmpfs           9.7G  182M  9.6G   2% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
efivarfs        128K   67K   57K  55% /sys/firmware/efi/efivars
tmpfs           9.7G  216K  9.7G   1% /run/qemu
/dev/sda1       511M  6.2M  505M   2% /boot/efi
tmpfs           2.0G  168K  2.0G   1% /run/user/1000
overlay         915G  319G  550G  37% /var/lib/docker/overlay2/ae062a270b32719d4
bd49ce602c42ebc68add1f290188002f21d9149529e7684/merged
/dev/loop3       32M   25M  7.6M  77% /home/ryan/tmp/P1_BOOT
/dev/loop24     107M   27M   73M  27% /home/ryan/tmp/P2_ROOTFS
```

2. 請準備好整個 SD card 備份及分別備份 SD card Partitions，若成功備份會類似如下輸出。

```
root@ryan-HPC:/home/ryan/Downloads/images# file /home/ryan/tmp/boot_part.img
/home/ryan/tmp/boot_part.img: DOS/MBR boot sector, code offset 0x58+2, OEM-ID "m
kfs.fat", sectors/cluster 8, Media descriptor 0xf8, sectors/track 32, heads 64,
hidden sectors 2048, sectors 1048576 (volumes > 32 MB), FAT (32 bit), sectors/FA
T 1024, reserved 0x1, serial number 0x78bd93c9, label: "BOOT       "
```

3. rsync backup 請列出 sys_backup 及 mmc2，內容應相同。

```
root@ryan-HPC:/mnt/sys_backup# ls /mnt/mmc2
bin   etc   lib64       lost+found  mnt  proc  run   sys  usr
dev   lib   linuxrc     media       opt  root  sbin  tmp  var
root@ryan-HPC:/mnt/sys_backup# ls /mnt/sys_backup/mmc2/
bin   etc   lib64       lost+found  mnt  proc  run   sys  usr
dev   lib   linuxrc     media       opt  root  sbin  tmp  var
```

4. 序列埠登入樹梅派。
5. 以上 partition 及 mount point 請在 demo 前先掛載完成，方便助教確認。